

PROJECT REPORT

PyPSA-Based Urban Microgrid Optimization

From a Simple 24-Hour Model to Four Advanced Tests

Dispatch • Capacity Planning • Transmission • Carbon Cost

Microgrid Project 26Summer

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Project Overview

This report follows five steps: one initial model and four advanced directions. It is a learning project, not a finished plan for a real city.

Abstract

I built a microgrid model in PyPSA and solved it with HiGHS. After a simple one-bus dispatch, I added Hefei renewable profiles, capacity planning, a two-bus case, and a carbon-tax test. The figures show how storage, transmission limits, and carbon cost change the result. These are model experiments, not forecasts.

Table 1. Project stages and the main question in each stage

Stage	Model size	Main question
Initial model	24 h, 1 bus	How do fixed generators and a battery meet hourly demand?
Direction 1: Hefei profiles	24 & 168 h, 1 bus	What changes when imported solar and wind profiles are used?
Direction 2: Capacity	24 h, 1 bus	What capacities are chosen when construction and land limits are added?
Direction 3: Transmission	24 h, 2 buses	What happens when renewable power must pass through a 15 MW line?
Direction 4: Carbon cost	24 h, 7 cases	How do capacity and emissions change as the carbon tax rises?

Reading note. Numbers come from the notebook and updated screenshots. The Hefei renewable profiles are data-based, but several load and cost inputs are representative assumptions.

1. Model Setup and Working Method

I used hourly time steps to show solar, wind, demand peaks, and battery charging. The solver chooses generator and battery output in every hour; in the planning cases it also chooses capacity. The objective is the lowest feasible model cost.

The main rules are simple:

- electricity supply must match the load in every hour;
- generator output cannot exceed installed capacity or the available renewable profile;
- battery energy moves between hours with 90% charging and 90% discharging efficiency; and
- the transmission case cannot send more than 15 MW through the line.

LMP in plain language. LMP is the extra model cost of one more MWh at a bus and hour. Two bus prices can differ when the line is full.

Battery sign. In the dispatch figures, positive battery power means discharge to the grid, while negative battery power means charging.

2. Initial 24-Hour Single-Bus Model

I started with a small case that could be checked by eye. Demand is 6-30 MW. The system has a 50 MW thermal unit at \$80/MWh, 30 MW of solar, 20 MW of wind, and a 10 MW four-hour battery. Solar and wind have zero marginal cost.

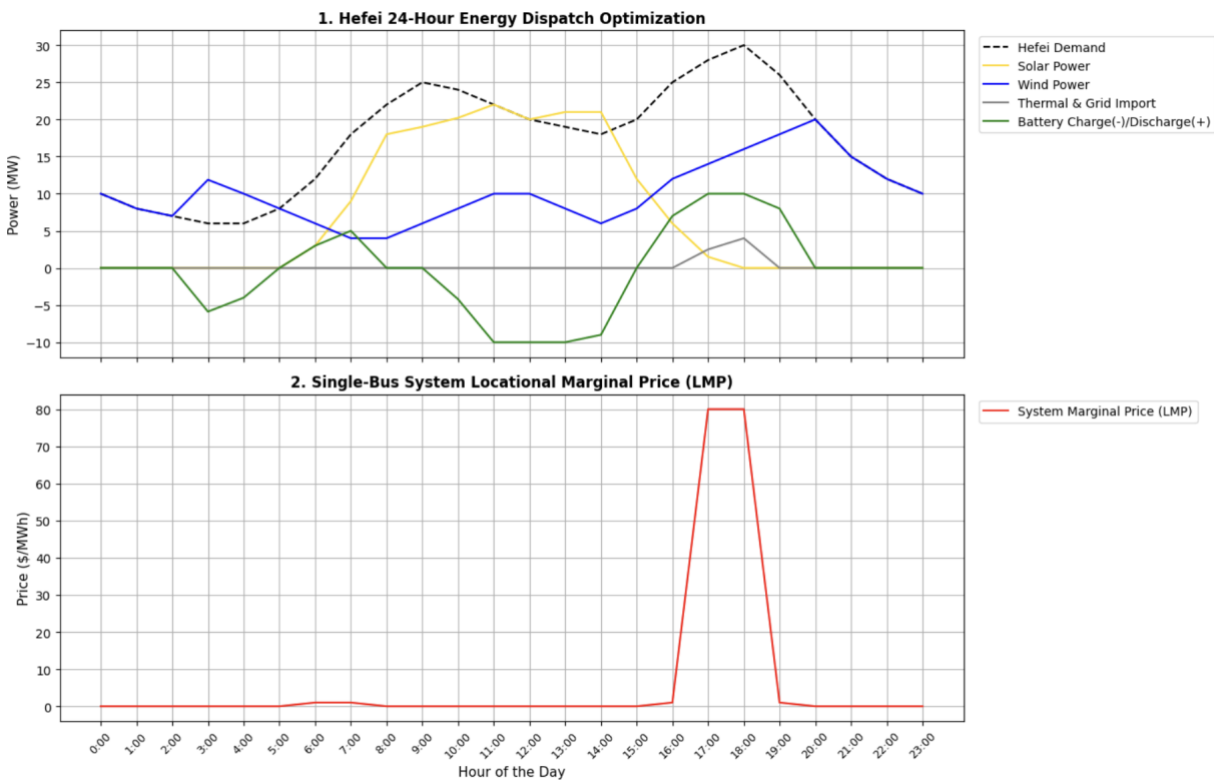


Figure 1. Initial 24-hour dispatch using synthetic teaching profiles and the resulting single-bus marginal price.

Renewables and the battery cover most hours. Around 17:00-18:00, high demand and falling solar require the thermal unit, so price reaches about \$80/MWh. At many other hours it is near zero or \$1/MWh. This checked the dispatch, battery sign, and price output.

Important label note. Although the chart title says “Hefei,” this initial example uses synthetic teaching profiles rather than measured Hefei data. The Hefei data-based extension begins in the next section.

3. Advanced Direction 1 - Hefei Renewable Profiles

This extension uses hourly Hefei solar and wind CSV profiles identified as Renewables.ninja data and shifted to China Standard Time. The representative system has 3,800-7,500 MW load, 2,000 MW solar, 300 MW wind, a 200 MW/800 MWh battery, and up to 8,000 MW of thermal generation or imports. Prices are \$30-\$115/MWh by time of use.

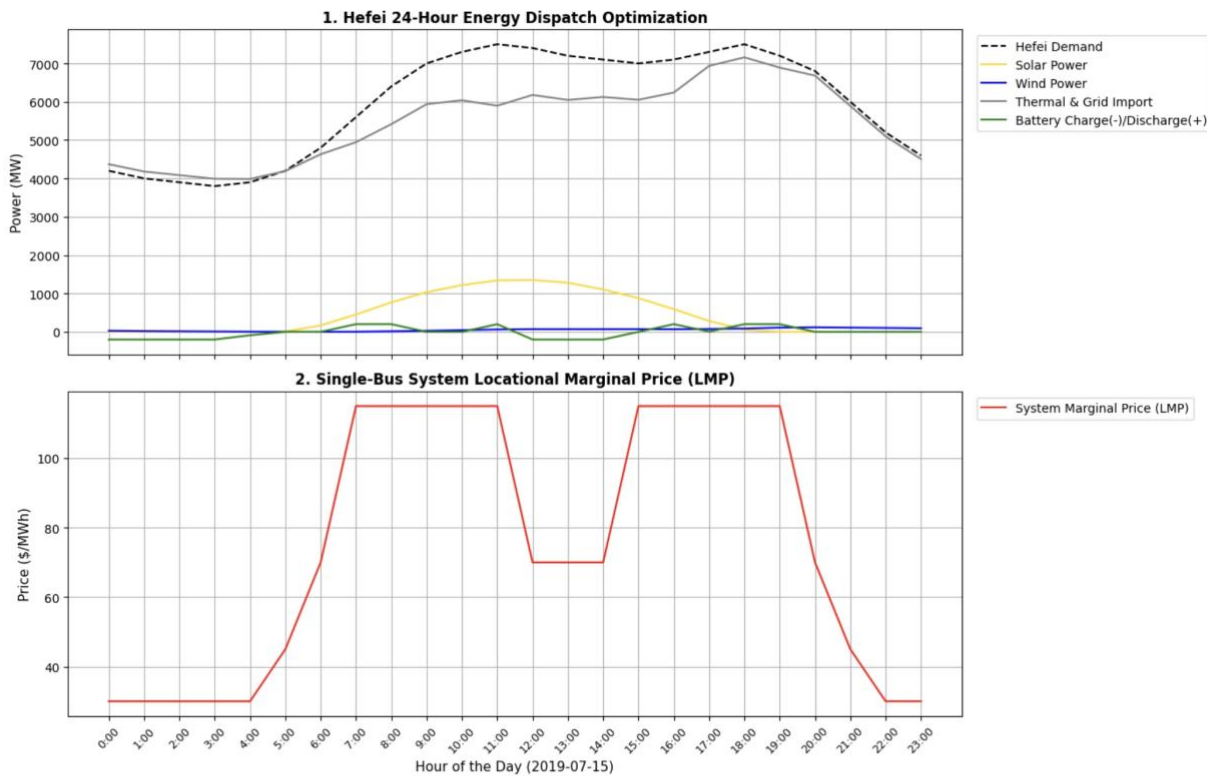


Figure 2. Hefei one-day dispatch, system price, and 168-hour extension.

On 15 July 2019, solar reaches about 1,350 MW at midday, but imports or thermal power still supply most demand. Reported costs are \$10,511,111.67 for the day and \$74,534,499.67 for 15-21 July. The week repeats daily load and price patterns, so it is a 168-hour model test rather than measured Hefei demand.

4. Advanced Direction 2 - Capacity Expansion with Land Limits

Here the solver chooses gas, solar, wind, and battery capacity. Solar is limited to 32 MW and wind to 26 MW as simple stand-ins for land or site limits. Construction and operating costs are considered together.

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1. Optimal Gas Capacity (Gas Cap) : 16.19 MW
2. Optimal Solar Capacity (Solar Cap) : 32.00 MW (Under Limit)
3. Optimal Wind Capacity (Wind Cap) : 8.04 MW (Under Limit)
4. Optimal Battery Capacity (Battery Cap): 13.41 MW
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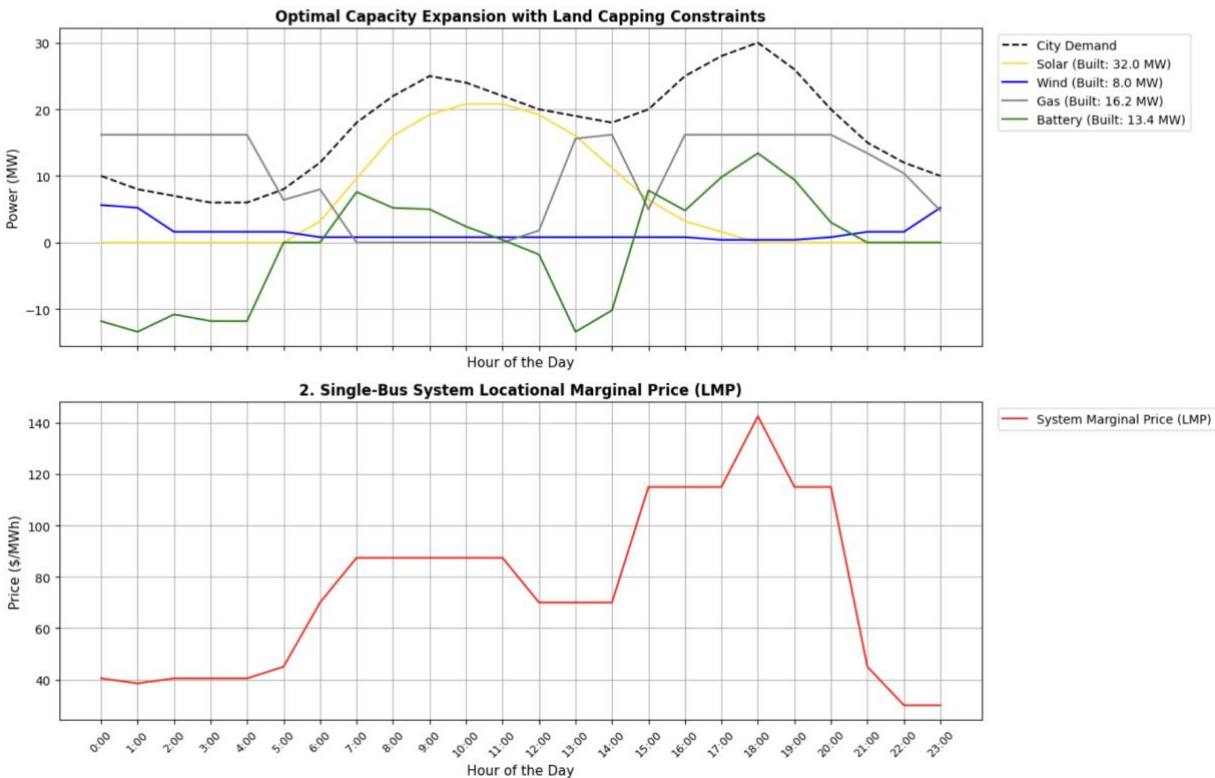


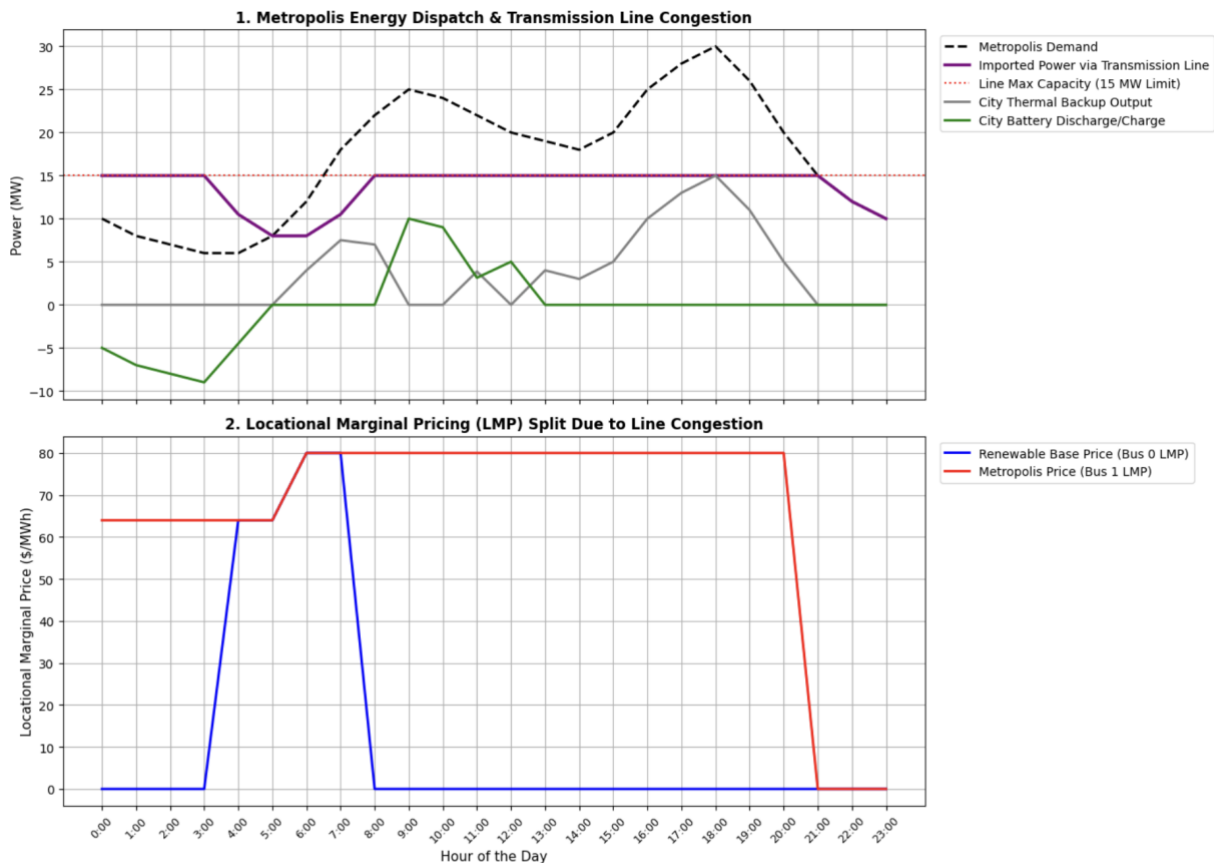
Figure 3. Capacity expansion result, hourly dispatch, and marginal price.

The model selects 16.19 MW gas, 32.00 MW solar, 8.04 MW wind, and 13.41 MW battery power. Solar reaches its limit, while wind does not. With these assumptions, solar plus storage is preferred, and price peaks during the tight evening hours.

Caution. The construction costs are simplified. These capacities compare model choices; they are not investment or land-use recommendations.

5. Advanced Direction 3 - Transmission Congestion

This case has two buses connected by a 15 MW line. The renewable base has 25 MW each of solar and wind plus 2 MW load. The city has 6-30 MW demand, a 50 MW thermal unit at \$80/MWh, and a 10 MW four-hour battery. The question is whether cheap power can reach the city.



The lowest total cost of a cross-regional dual-node power grid throughout the day: \$7,096.34 USD

Figure 4. Two-bus dispatch, 15 MW line congestion, and LMP separation.

The line reaches 15 MW in many hours, so local thermal power runs even when renewable power is available at the other bus. The base price can be near zero while the city reaches \$80/MWh. Daily cost is \$7,096.34, showing why line capacity matters.

Model lesson. Storage moves energy between hours, while transmission moves energy between places. A system may need both kinds of flexibility.

6. Advanced Direction 4 - Carbon Tax Sensitivity

The final test repeats capacity planning for carbon taxes from \$0 to \$150/tCO₂ in \$25 steps. Gas is assigned 0.9 tCO₂/MWh, so each tax raises its operating cost before the model chooses capacity again.

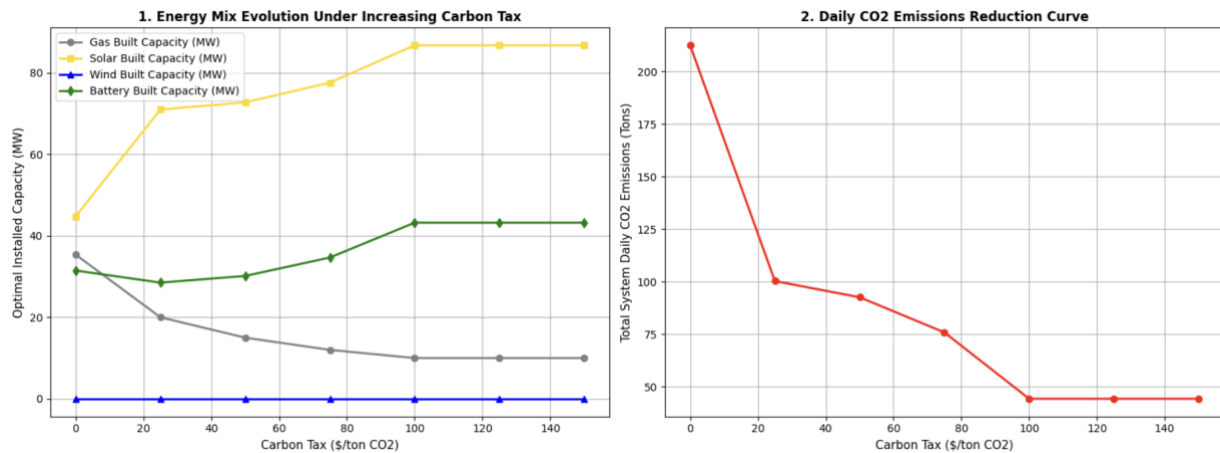


Figure 5. Capacity choices and daily CO₂ emissions as the carbon tax increases.

From \$0 to \$100/tCO₂, gas falls from about 35 MW to 10 MW, solar rises from 45 MW to 87 MW, and battery power rises from 31 MW to 43 MW. Wind stays at zero under these cost assumptions. Daily emissions fall from about 212 to 44 tons, then change little above \$100/tCO₂.

\$100/tCO₂ is only this model's turning point, not a policy answer. A real study needs annual costs, fuel prices, more technologies, and a longer period.

7. What Changed from One Stage to the Next

Each stage added one question. The initial model checked basic operation. The Hefei case used better renewable time series and a longer run. Capacity expansion asked what to build, the two-bus case added location and a line limit, and the carbon test changed the cost of using gas.

Main Lessons

- Low operating cost makes solar and wind attractive, but their timing still matters.
- A battery can reduce a timing mismatch, although it loses some energy while charging and discharging.
- A transmission limit can keep cheap power away from a load center and create different local prices.
- Capacity results can change a lot when technology costs, land limits, or carbon costs change.

8. Limitations and Next Steps

This is a learning model. Some load, price, and capacity inputs are representative, and the weekly case repeats one load profile. Investment costs are not on one annual basis, so section costs should not be compared directly. The network has only one or two buses, and the notebook warns about undefined carriers and zero-line resistance.

The next version could focus on a few practical improvements:

- use matching 8,760-hour load, solar, wind, and fuel-price data;
- convert all construction and operating costs to clear, consistent units and years;
- plot battery state of charge in addition to charging and discharging power;
- expand the network to at least three regional buses and test line expansion.

9. Conclusion

The five stages cover basic dispatch, Hefei profiles, capacity planning, transmission, and carbon cost. My main lesson is that renewable capacity is only one part of the system. Storage moves electricity through time, transmission moves it between places, and costs shape both choices. Better data and a full-year run would make a stronger next project.

Source and Model Notes

Complete source code and reproducible notebook are available in the project repository:
github.com/kevdb1/microgrid-energy-optimization (notebook: Microgrid_Project_26Summer.ipynb).

Software used in the notebook: PyPSA, HiGHS, pandas, NumPy, and Matplotlib.

Hefei solar and wind CSV files are identified in the notebook as Renewables.ninja data and shifted from UTC to China Standard Time.

All figures and numerical results were generated directly from this notebook.